

Assignment 7

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Problem Statement

To develop a text preprocessing and analysis application using NLTK for performing tokenization, part-of-speech (POS) tagging, and basic Natural Language Processing tasks.

Objective

The objective of this assignment is to understand and implement fundamental NLP preprocessing techniques such as tokenization, stop word removal, POS tagging, and frequency analysis using the NLTK library.

Terminologies Used

1. Tokenization

Tokenization is the process of splitting text into smaller units such as sentences or words. It is the first step in text preprocessing.

2. Sentence Tokenization

Sentence tokenization divides a paragraph into individual sentences.

3. Word Tokenization

Word tokenization splits sentences into individual words or tokens.

4. Stop Words

Stop words are commonly used words such as "is", "the", and "and" that do not add significant meaning to the text and are often removed during preprocessing.

5. Part-of-Speech (POS) Tagging

POS tagging assigns grammatical categories such as noun, verb, adjective, etc., to each word in a sentence.

6. Frequency Distribution

Frequency distribution measures how often each word appears in the text. It helps in identifying important or commonly used words.

Methodology

The following steps were performed:

1. Text data was provided as input.
 2. Sentence tokenization was applied to split the text into sentences.
 3. Word tokenization was performed to extract individual words.
 4. Stop words were removed to retain meaningful words.
 5. POS tagging was applied to identify grammatical roles of words.
 6. Frequency distribution was calculated to analyze word occurrence.
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Output Description

- **Sentences:** List of sentences obtained after sentence tokenization.
 - **Tokens:** List of words obtained after word tokenization.
 - **Filtered Words:** Words after removing stop words.
 - **POS Tags:** Words along with their grammatical categories.
 - **Frequency Distribution:** Most frequent words in the text.
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Applications

- Text classification
 - Sentiment analysis
 - Chatbots and virtual assistants
 - Information retrieval
 - Text summarization
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Conclusion

The assignment demonstrates how basic NLP preprocessing and analysis tasks can be performed using NLTK. Tokenization, stop word removal, POS tagging, and frequency analysis help in transforming raw text into structured and meaningful data.

Result

A text preprocessing and analysis application was successfully implemented using NLTK, fulfilling the objectives of the assignment.